

CS471-lab11

Task 1: Student CRUD with ForeignKey (One Address per Student)

[forms.py](#) :

```
from .models import Book, Student, Address

class StudentForm(forms.ModelForm):
    class Meta:
        model = Student
        fields = ['name', 'age', 'address']
```

[views.py](#) :

```
from .models import Book, Publisher, Author, Address, Student

def lab11_students(request):
    students = Student.objects.all()
    return render(request, 'bookmodule/lab11_students.html', {'students': students})

def lab11_addstudent(request):
    if request.method == 'POST':
        form = StudentForm(request.POST)
        if form.is_valid():
            form.save()
            return redirect('/books/lab11/students')
    else:
        form = StudentForm()
    return render(request, 'bookmodule/lab11_addstudent.html', {'form': form})

def lab11_editstudent(request, id):
    student = Student.objects.get(id=id)
    if request.method == 'POST':
        form = StudentForm(request.POST, instance=student)
        if form.is_valid():
            form.save()
            return redirect('/books/lab11/students')
    else:
        form = StudentForm(instance=student)
    return render(request, 'bookmodule/lab11_editstudent.html', {'form': form})

def lab11_deletestudent(request, id):
    student = Student.objects.get(id=id)
    student.delete()
    return redirect('/books/lab11/students')
```

```
path('lab11/students', views.lab11_students),
path('lab11/addstudent', views.lab11_addstudent),
path('lab11/editstudent/<int:id>', views.lab11_editstudent),
path('lab11/deletestudent/<int:id>', views.lab11_deletestudent),
```

← ↻ ⓘ 127.0.0.1:8000/books/lab11/students

Home List Books Search

Student List

[+ Add New Student](#)

ID	Name	Age	Address	Actions
1	layan	20	jeddah	Edit Delete
4	lana	30	buraydah	Edit Delete
5	layan2	22	riyadh	Edit Delete

← ↻ ⓘ 127.0.0.1:8000/books/lab11/addstudent

Home List Books Search

Add New Student

Name:

Age:

Address:

[Cancel](#)

[←](#) [↺](#) [127.0.0.1:8000/books/lab11/editstudent/1](#)

[Home](#) [List Books](#) [Search](#)

Edit Student

Name:

Age:

Address:

[Cancel](#)

Task 2: Many-to-Many Relationship

[models.py](#):

```
class Address2(models.Model):
    city = models.CharField(max_length=100)

    def __str__(self):
        return self.city

class Student2(models.Model):
    name = models.CharField(max_length=100)
    age = models.IntegerField(default=0)
    addresses = models.ManyToManyField(Address2)

    def __str__(self):
        return self.name
```

[forms.py](#):

```

from .models import Book, Student, Address, Student2, Address2

class Student2Form(forms.ModelForm):
    class Meta:
        model = Student2
        fields = ['name', 'age', 'addresses']
        widgets = {
            'addresses': forms.CheckboxSelectMultiple()
        }

```

views.py:

```

def lab11_students2(request):
    students = Student2.objects.all()
    return render(request, 'bookmodule/lab11_students2.html', {'students': students})

def lab11_addstudent2(request):
    if request.method == 'POST':
        form = Student2Form(request.POST)
        if form.is_valid():
            form.save()
            return redirect('/books/lab11/students2')
    else:
        form = Student2Form()
    return render(request, 'bookmodule/lab11_addstudent2.html', {'form': form})

def lab11_editstudent2(request, id):
    student = Student2.objects.get(id=id)
    if request.method == 'POST':
        form = Student2Form(request.POST, instance=student)
        if form.is_valid():
            form.save()
            return redirect('/books/lab11/students2')
    else:
        form = Student2Form(instance=student)
    return render(request, 'bookmodule/lab11_editstudent2.html', {'form': form})

def lab11_deletestudent2(request, id):
    student = Student2.objects.get(id=id)
    student.delete()
    return redirect('/books/lab11/students2')

```

urls.py:

```

path('lab11/students2', views.lab11_students2),
path('lab11/addstudent2', views.lab11_addstudent2),
path('lab11/editstudent2/<int:id>', views.lab11_editstudent2),
path('lab11/deletestudent2/<int:id>', views.lab11_deletestudent2),

```

Student List (Many Addresses)

[+ Add New Student](#)

ID	Name	Age	Addresses	Actions
1	Abeer	27	Riyadh, Jeddah	Edit Delete
2	lamar	30	Dammam, Buraydah	Edit Delete
4	nora	23	Jeddah, Dammam	Edit Delete



127.0.0.1:8000/books/lab11/addstudent2

[Home](#)

[List Books](#)

[Search](#)

Add New Student (Many Addresses)

Name:

Age:

Addresses:

- ☐ Riyadh
- ☒ Jeddah
- ☒ Dammam
- ☒ Mecca
- ☐ Buraydah

[Cancel](#)

[←](#) [↻](#) [i](#) 127.0.0.1:8000/books/lab11/editstudent2/2

[Home](#) [List Books](#) [Search](#)

Edit Student (Many Addresses)

Name:

Age:

Addresses:

- ☐ Riyadh
- ☐ Jeddah
- ☒ Dammam
- ☐ Mecca
- ☒ Buraydah

[Cancel](#)

Task 3: Image Upload

[settings.py](#):

```
import os
MEDIA_URL = '/media/'
MEDIA_ROOT = os.path.join(BASE_DIR, 'media')
```

main [urls.py](#):

```
from django.conf import settings
from django.conf.urls.static import static

urlpatterns = [
    path('admin/', admin.site.urls),
    path('books/', include('apps.bookmodule.urls')),
] + static(settings.MEDIA_URL, document_root=settings.MEDIA_ROOT)
```

[models.py](#):

```
class Product(models.Model):
    name = models.CharField(max_length=100)
    description = models.TextField()
    image = models.ImageField(upload_to='products/')

    def __str__(self):
        return self.name
```

forms.py:

```
from .models import Book, Student, Address, Student2, Address2, Product

class ProductForm(forms.ModelForm):
    class Meta:
        model = Product
        fields = ['name', 'description', 'image']
```

views.py:

```
from .forms import BookForm, StudentForm, Student2Form, ProductForm

def lab11_products(request):
    products = Product.objects.all()
    return render(request, 'bookmodule/lab11_products.html', {'products': products})

def lab11_addproduct(request):
    if request.method == 'POST':
        form = ProductForm(request.POST, request.FILES)
        if form.is_valid():
            form.save()
            return redirect('/books/lab11/products')
    else:
        form = ProductForm()
    return render(request, 'bookmodule/lab11_addproduct.html', {'form': form})

def lab11_editproduct(request, id):
    product = Product.objects.get(id=id)
    if request.method == 'POST':
        form = ProductForm(request.POST, request.FILES, instance=product)
        if form.is_valid():
            form.save()
            return redirect('/books/lab11/products')
    else:
        form = ProductForm(instance=product)
    return render(request, 'bookmodule/lab11_editproduct.html', {'form': form, 'product': product})

def lab11_deleteproduct(request, id):
    product = Product.objects.get(id=id)
    product.delete()
    return redirect('/books/lab11/products')
```

[urls.py](#):


```

path('lab11/products', views.lab11_products),
path('lab11/addproduct', views.lab11_addproduct),
path('lab11/editproduct/<int:id>', views.lab11_editproduct),
path('lab11/deleteproduct/<int:id>', views.lab11_deleteproduct),

```

←

→

↺

i

127.0.0.1:8000/books/lab11/products

Home

List Books

Search

Product List

[+ Add New Product](#)

ID	Name	Description	Image	Actions
1	book	this is a new book		Edit Delete
2	book4	this is book4		Edit Delete
4	book5	this is book5		Edit Delete

Add New Product

Name:

Description:

this is book 6

Image: Screenshot (19).png

[Cancel](#)

Edit Product

Name:

this is book4

Description:

Image: Currently: [products/Screenshot_2026-01-09_220235.png](#)

Change: No file chosen



Current image:

[Cancel](#)